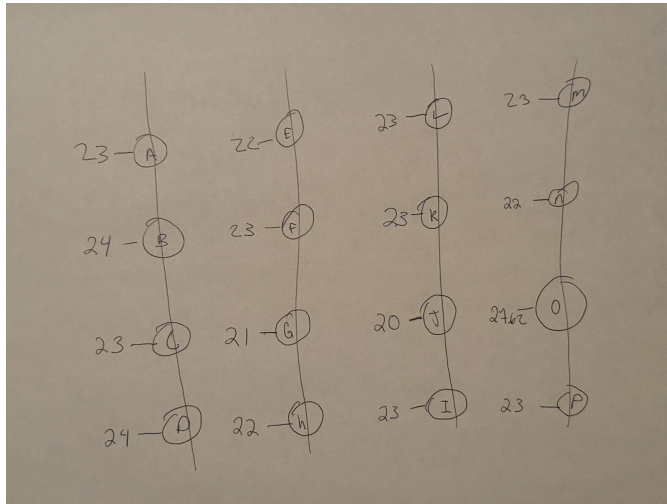


Distributed Observation and Checkpoints



Preamble

This document exists in parallel with the Harding Systems Design Manifesto to provide contextual background on how its principles were formed. It is not required reading, and it does not modify or justify the manifesto. The principles stand independently. What follows is simply an account of how a particular way of seeing systems took shape.

Note

One of my earliest and most enduring influences came from a scene in the film *Battleship*, where a constrained naval force, whose default tracking systems were all blocked by the adversaries, infers the position of an unseen adversary using a field of passive buoys. Each buoy reports only local disturbance. No single sensor possesses global truth. Meaning emerges only through correlation across space and time. **Silence is as informative as signal.** Interpretation remains human.

That scene stayed with me not because of its spectacle, but because of its epistemology. The system does not pretend to see what it cannot. It does not mask uncertainty. It places honest observers in the environment and accepts that truth must be inferred, not asserted.

Over time, this model quietly shaped how I approached real systems. In production environments, I began treating processes as checkpoints rather than authorities. Each station, sensor, or step reports what it locally knows — no more, no less. Missing signals matter. Gaps matter. Downstream compensation is treated as a risk, not a solution, because it erases causality. When one checkpoint absorbs the responsibility of another, the system loses its ability to learn.

This thinking directly informed systems such as component tracking pipelines, where the movement of parts through production is inferred by their appearance — or absence — at defined points, rather than by centralized polling or retrospective reconciliation. It also influenced environmental and volatility monitoring systems, where conditions are observed through distributed measurement rather than averaged away at a single control point.

What matters in all of these designs is not optimization, speed, or appearance of continuity. What matters is that the system remains truthful about what it can and cannot see. Failure is allowed to surface. Context is preserved. Interpretation remains explicit rather than hidden inside automation.

The manifesto grew out of repeatedly encountering environments where these ideas were under pressure — where there was incentive to smooth over gaps, mask upstream failure, or reassign responsibility to preserve throughput. Each time, the same discomfort appeared: systems that hide their own uncertainty become fragile, even when they appear to function.

The principles articulated in the manifesto are an attempt to make that discomfort explicit and actionable. They are not a reaction to a single system or organization, but a commitment to designing systems that can survive disagreement, constraint, and time without losing their ability to tell the truth.

This origin story is included only to illuminate how that commitment formed. The manifesto itself does not depend on it.